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COMPUTER NETWORKS

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Department of Computer Science and Engineering

Unit – 4 Network Layer and Internet Protocol

4.5 Introduction to Network Layer Protocols

- DHCP
- ICMP

4.6 IPv6 Addressing

4.7 Introduction to Routing Algorithms

- 128 bits address
- 340 undecillion address (340 + 33 digits)
- IPv6 address space is 2^{96} times of the IPv4 address.
- Made up of eight 16-bit blocks (octet)
- Separated by ‘:’
- Hexadecimal, each hexadecimal character – 4 bits

340,282,366,920,938,463,374,607,431,768,211,456

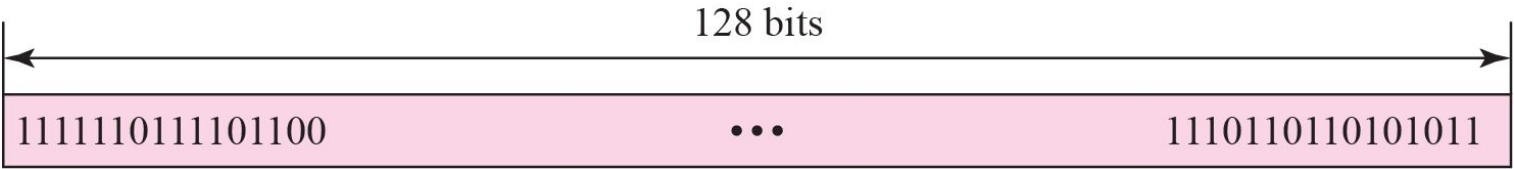
Rules:

- Discard leading Zero(es)
- If two or more blocks contain consecutive Zero(es), omit them all and replace with double colon sign ::
- If Zero(es) are present in a discontinuous pattern, replace them with :: at only one joining.

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

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Introduction to IPv6



Colon hexadecimal notation FDEC : BA98 : 7654 : 3210 : ADBF : BBFF : 2922 : FFFF

Zero compression



CIDR address



Rule 1

2001:0000:3238:DFE1:0063:0000:0000:FEFB

2001:0000:3238:DFE1:63:0000:0000:FEFB

Rule 2

2001:0000:3238:DFE1:0063:0000:0000:FEFB

2001:0000:3238:DFE1:63::FEFB
2001:0:3238:DFE1:63::FEFB

Show the unabbreviated colon hex notation for the following IPv6 addresses:

- a. An address with 64 0s followed by 64 1s.
- b. An address with 128 0s.
- c. An address with 128 1s.
- d. An address with 128 alternative 1s and 0s.

Solution

- a. 0000:0000:0000:0000:FFFF:FFFF:FFFF:FFFF
- b. 0000:0000:0000:0000:0000:0000:0000:0000
- c. FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF
- d. AAAA:AAAA:AAAA:AAAA:AAAA:AAAA:AAAA:AAAA



The following shows the zero contraction version of addresses in previous example.

- a. :: FFFF:FFFF:FFFF:FFFF
- b. ::
- c. FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF
- d. AAAA:AAAA:AAAA:AAAA:AAAA:AAAA:AAAA:AAAA



Show abbreviations for the following addresses:

- a. 0000:0000:FFFF:0000:0000:0000:0000:0000
- b. 1234:2346:0000:0000:0000:0000:0000:1111
- c. 0000:0001:0000:0000:0000:0000:1200:1000
- d. 0000:0000:0000:0000:0000:FFFF:24.123.12.6

Solution

- a. 0:0:FFFF::
- b. 1234:2346::1111
- c. 0:1::1200:1000
- d. ::FFFF:24.123.12.6



Decompress the following addresses and show the complete unabbreviated IPv6 address:

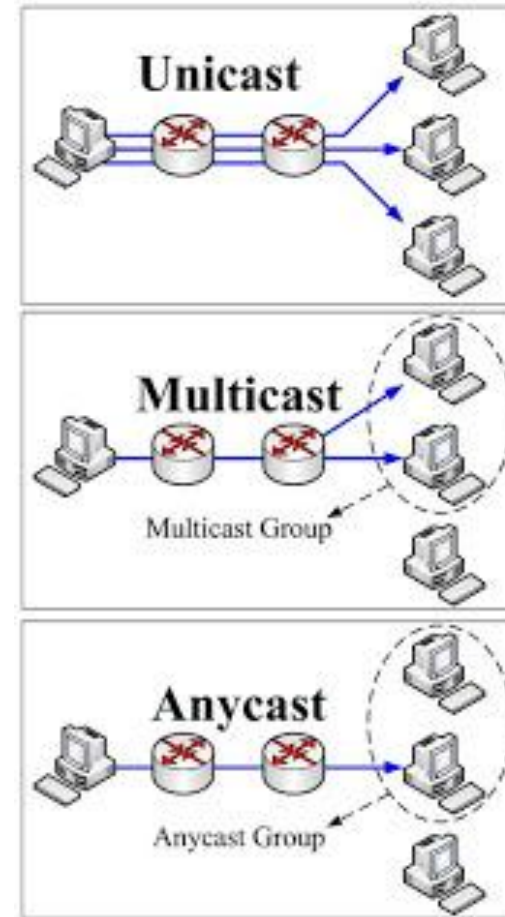
- a. 1111::2222
- b. ::
- c. 0:1::
- d. AAAA:A:AA::1234

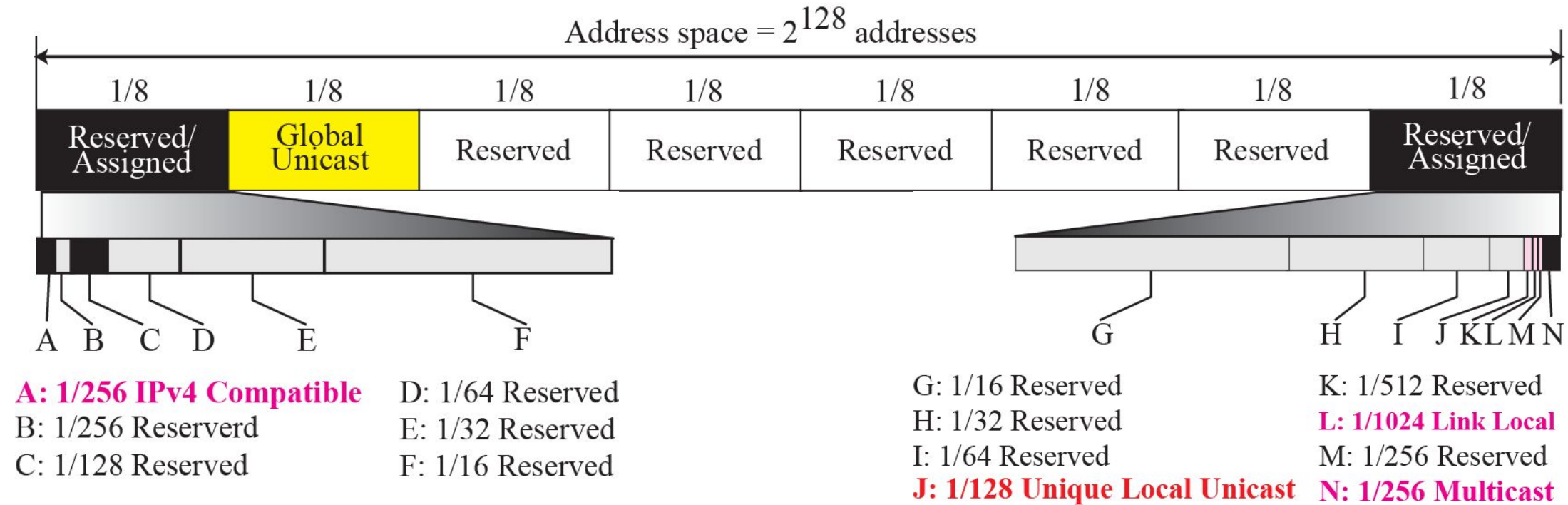
Solution

- a. 1111:0000:0000:0000:0000:0000:0000:2222
- b. 0000:0000:0000:0000:0000:0000:0000:0000
- c. 0000:0001:0000:0000:0000:0000:0000:0000
- d. AAAA:000A:00AA:0000:0000:0000:0000:1234



- In IPv6, a destination address belong to:
 - **Unicast Address (one-to-one)** – defines a single interface (computer or router)
 - **Multicast Address (one-to-many)** – also defines a group of computers, each member of the group receives a copy.
 - **Anycast Address (one-to-nearest)** – defines a group of computers that all share a single address.
 - **Broadcast Address (one-to-all)** – ×





- Larger address space
- Better header format
- New options
- Allowance for extension
- Support for resource allocation
- Support for more security

checksum: removed entirely to reduce processing time at each hop

options: allowed, but outside of header, indicated by “Next Header” field

ICMPv6: new version of ICMP

fragmentation/reassembly: no

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IPv6 Datagram Format

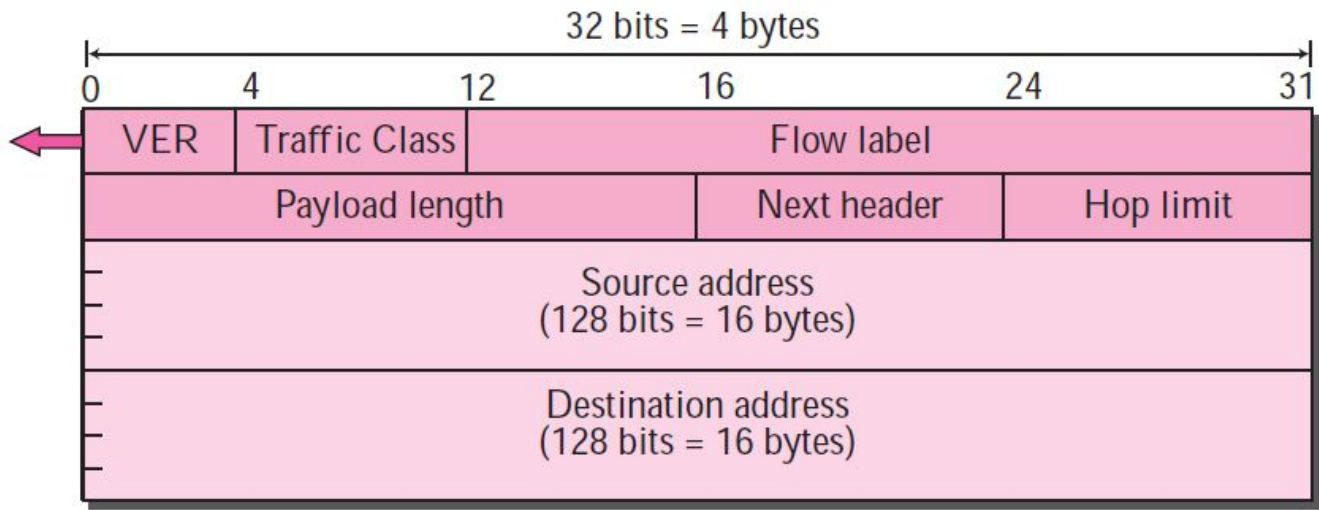
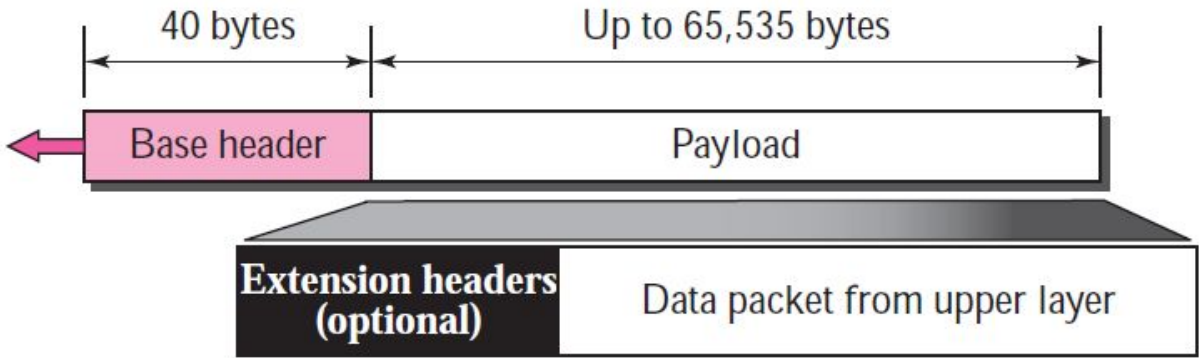
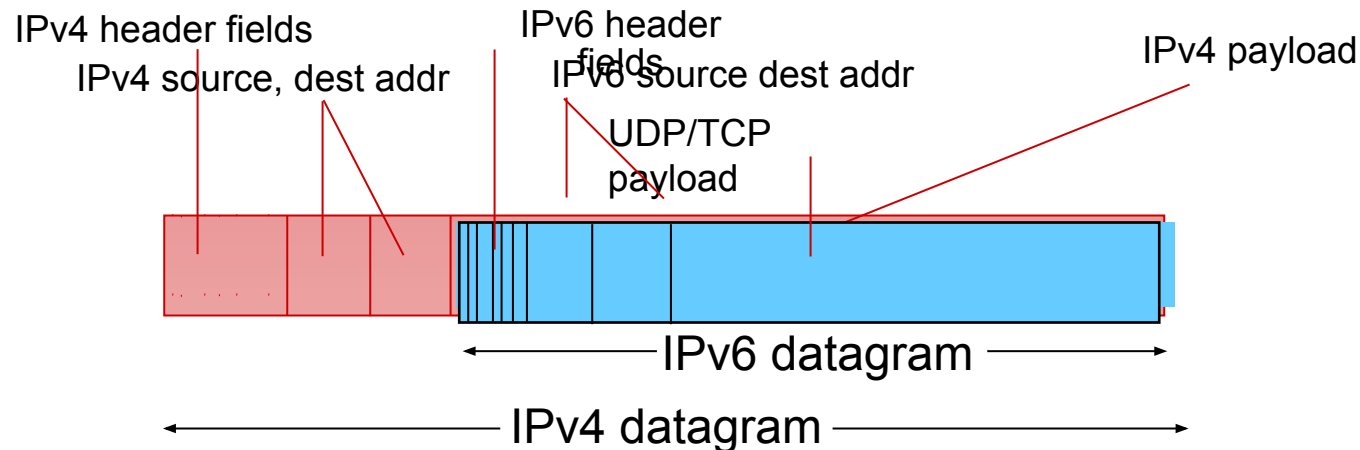


Table 27.1 Next Header Codes

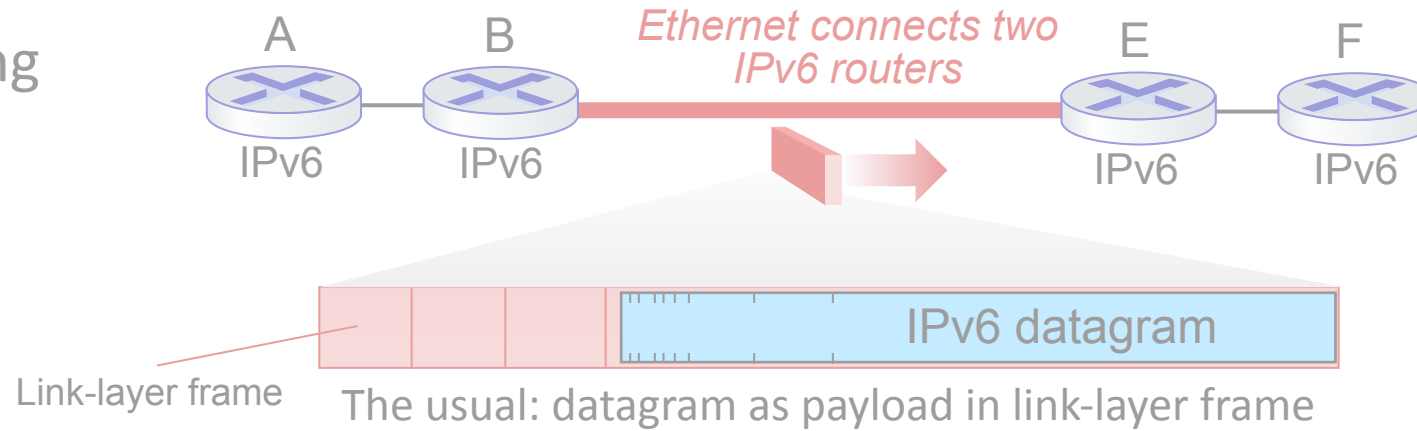
Code	Next Header	Code	Next Header
0	Hop-by-hop option	44	Fragmentation
2	ICMP	50	Encrypted security payload
6	TCP	51	Authentication
17	UDP	59	Null (No next header)
43	Source routing	60	Destination option

Format of the base header

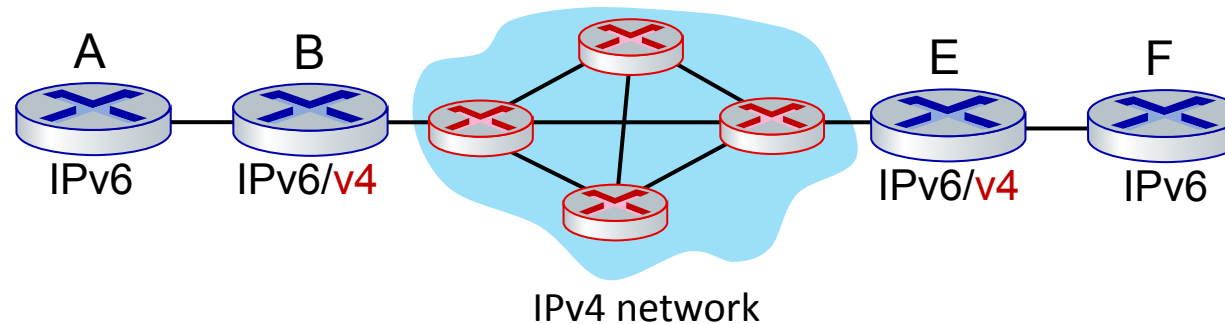
- not all routers can be upgraded simultaneously
 - no “flag days”
 - how will network operate with mixed IPv4 and IPv6 routers?
- **tunneling**: IPv6 datagram carried as *payload* in IPv4 datagram among IPv4 routers (“packet within a packet”)
 - tunneling used extensively in other contexts (4G/5G)



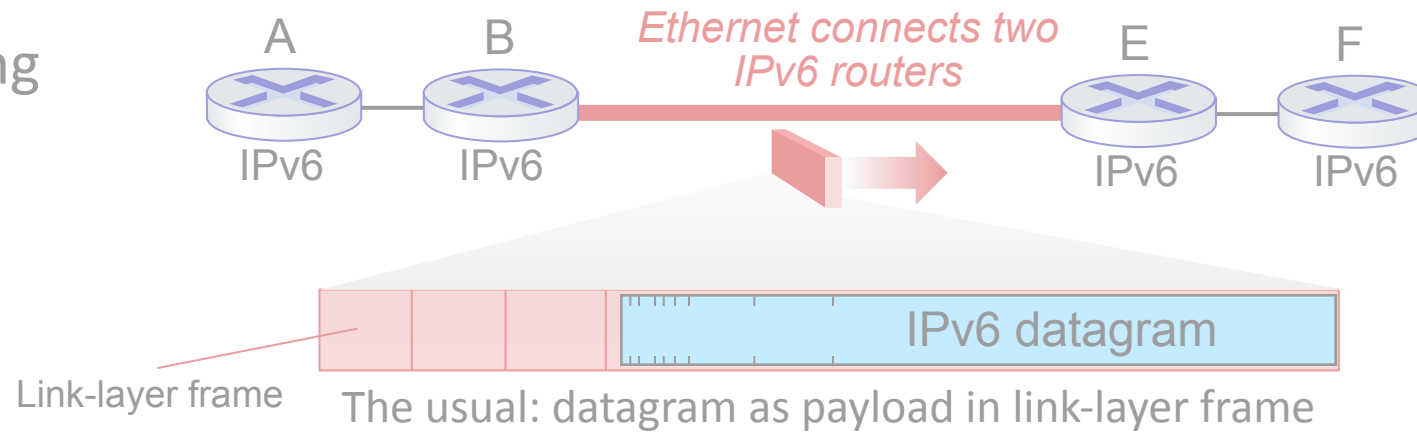
Ethernet connecting
two IPv6 routers:



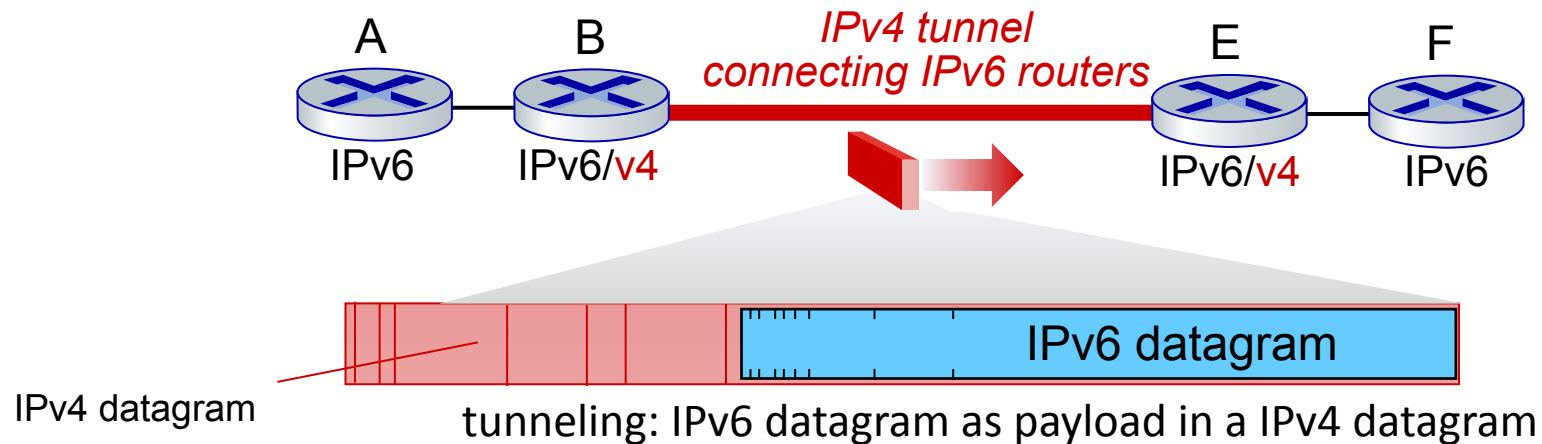
IPv4 network
connecting two
IPv6 routers

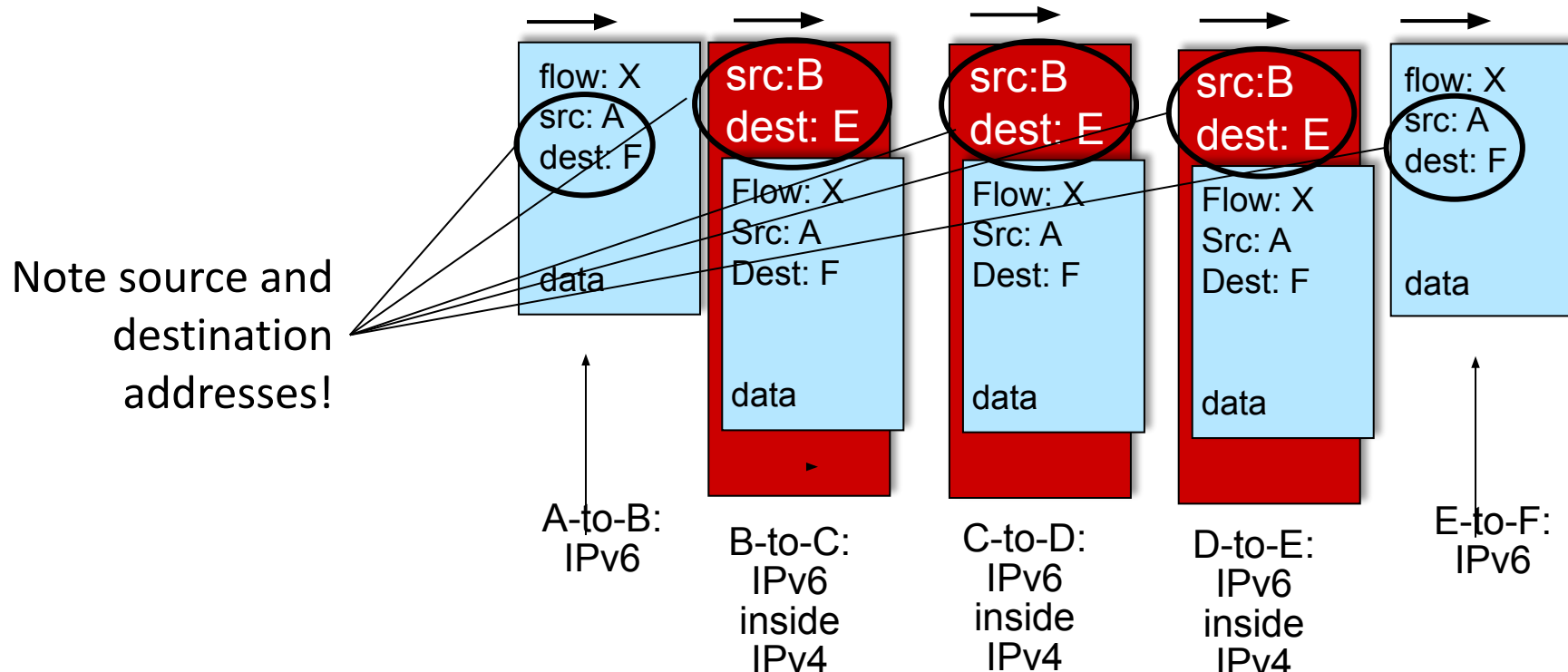
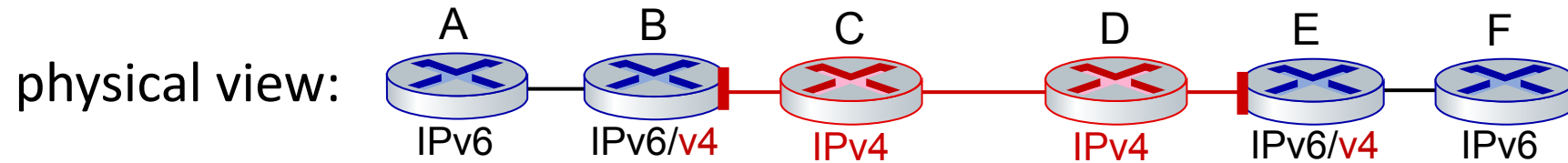
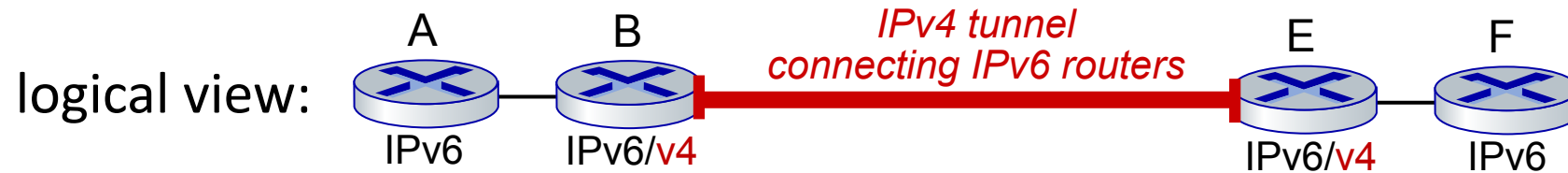


Ethernet connecting
two IPv6 routers:



IPv4 tunnel
connecting two
IPv6 routers





<https://www.akamai.com/us/en/resources/our-thinking/state-of-the-internet-report/state-of-the-internet-ipv6-adoption-visualization.jsp>



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THANK YOU

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